THE LANGUAGE OF

MATHEMATICS

General notation

Symbol	Meaning	Examples
$=$	Equal	$2+3=5$; $4-4=0$
$\neq$	Not equal	$2\neq 3$; $-1\neq 1$
$\approx$	Approximately	$\pi\approx 3.14$ $\sqrt{2}\approx 1.41$
$<$, $>$	Less than; greater than	$2<3$; $-1<0$ $5>4$; $0>-2$
$\leq$, $\geq$	Less than or equal to; greater than or equal to	$2\leq 2$; $-1\leq 3$ $5\geq 4$; $0\geq 0$
$\pm$	Plus or minus	$3\pm 1=4$ or 2 $-2\pm 3=1$ or -5
$\lvert a\rvert$	Absolute value	$\lvert -5\rvert=5$; $\lvert 3\rvert=3$
$\sqrt{a}$	Square root	$\sqrt{9}=3$; $\sqrt{\frac{1}{4}}=\frac{1}{2}$

Sets and number systems

Symbol	Meaning	Examples
$\{a,b,c\}$	Set with three elements	$\{2,4,6\}$; $\{-1,0,5\}$
$\varnothing$	Empty set	$\{1\}\cap\{2\}=\varnothing$ $\{x\in\mathbb{R}: x^2=-1\}=\varnothing$
$\in$	Is element of	$2\in\{1,2\}$; $-1\in\mathbb{Z}$
$\notin$	Is not element of	$3\notin\{1,2\}$; $\frac{1}{2}\notin\mathbb{Z}$
$\subseteq$	Is a subset of	$\{1\}\subseteq\{1,2\}$ $\mathbb{N}\subseteq\mathbb{Z}$
$\cup$, $\cap$	Union; intersection	$\{1\}\cup\{2\}=\{1,2\}$ $\varnothing\cup\{0\}=\{0\}$ $\{1,2\}\cap\{2\}=\{2\}$ $\{-1\}\cap\{4\}=\varnothing$
$\mathbb{N}$, $\mathbb{Z}$	Natural numbers; integers	$3\in\mathbb{N}$; $8\in\mathbb{N}$ $-2\in\mathbb{Z}$; $0\in\mathbb{Z}$
$\mathbb{Q}$, $\mathbb{R}$	Rational numbers; real numbers	$\frac{1}{2}\in\mathbb{Q}$; $-3\in\mathbb{Q}$ $\sqrt{2}\in\mathbb{R}$; $\frac{1}{4}\in\mathbb{R}$
(a,b), $[a,b]$	Open interval; closed interval	$1\notin(1,3)$; $0\in(-2,2)$ $1\in[1,3]$; $-2\in[-2,0]$

Summation notation

Symbol	Meaning	Examples
$\sum_{i=1}^{n} a_i$	Add a_i for each integer i from 1 to n.	$\sum_{i=1}^{3} 2i=2+4+6$ $\sum_{i=1}^{2}(i+3)=4+5$

In $\sum_{i=1}^{n} a_i$, i is the **index**, 1 is the **lower bound**, n is the **upper bound**, and a_i is the **expression being summed**.

Worked examples

$\sum_{i=1}^{4} i=1+2+3+4=10$

$\sum_{k=1}^{3} k^2=1+4+9=14$

Probability and events

Example contexts
I: fair die **II**: two independent fair coin tosses.
H = heads; T = tails; letters give the first, then second toss.
II: $A=\{\text{HH}\}$: two heads;
$B=\{\text{HH},\text{HT},\text{TH}\}$: at least one head.

Symbol	Meaning	Examples
Ω	Sample space: all possible outcomes	I: $\Omega=\{1,\ldots,6\}$ II: $\Omega=\{\text{HH},\text{HT},\text{TH},\text{TT}\}$
ω	An individual outcome	I: $\omega=3\in\Omega$ II: $\omega=\text{HT}\in\Omega$
$A=\{1,3,5\}$	Event shown as a set of outcomes	I: $A=\{1,3,5\}$: odd II: $A=\{\text{HH}\}$: two heads
$A\subseteq\Omega$	An event is a subset of the sample space	I: $\{1,3,5\}\subseteq\Omega$ II: $A\subseteq\Omega$
$P(A)$	Probability of event A	I: $P(A)=\frac{3}{6}=\frac{1}{2}$ II: $P(A)=\frac{1}{4}$
A^c	Complement of event A	I: $A^c=\{2,4,6\}$ II: $A^c=\{\text{HT},\text{TH},\text{TT}\}$
$A\cup B$	Event A or B (or both)	I: $A\cup B=\{1,2,3,5\}$ II: $A\cup B=\{\text{HH},\text{HT},\text{TH}\}$
$A\cap B$	Both events A and B	I: $A\cap B=\{1\}$ II: $A\cap B=\{\text{HH}\}$
$P(A\mid B)$	Probability of A given B	I: $P(A\mid B)=\frac{1}{2}$ II: $P(A\mid B)=\frac{1}{3}$

Function notation

Example contexts
I: $f:\mathbb{R}\to\mathbb{R}$, $f(x)=x^2$ **II**: $f:[0,\infty)\to\mathbb{R}$, $f(x)=\sqrt{x}$

Symbol	Meaning	Examples
f	Name of a function	I: $f(x)=x^2$ II: $f(x)=\sqrt{x}$
$f(x)$	Value of f at input x	I: $f(3)=3^2=9$ II: $f(4)=\sqrt{4}=2$
$f:A\to B$	Function from set A to set B	I: $f:\mathbb{R}\to\mathbb{R}$ II: $f:[0,\infty)\to\mathbb{R}$
$x\mapsto f(x)$	Input-to-output mapping	I: $x\mapsto x^2$; $3\mapsto 9$ II: $x\mapsto\sqrt{x}$; $4\mapsto 2$
$\text{dom}(f)$	Domain of a function	I: $\text{dom}(f)=\mathbb{R}$ II: $\text{dom}(f)=[0,\infty)$

Letters and Greek symbols

Symbol	Name / description
$a,b,c,\ldots$	Common lowercase Latin letters
$A,B,C,\ldots$	Common uppercase Latin letters
x,y,z	Other common Latin letters
α, β, γ	Alpha; beta; gamma
δ, Δ	Delta; uppercase delta
θ, Θ	Theta; uppercase theta
λ, Λ	Lambda; uppercase lambda
μ, σ	Mu; sigma
ω, Ω	Omega; uppercase omega

Variable Examples

A variable is a symbol that can represent different values. Its allowed values and meaning depend on the context.

Possible values in separate cases

Each choice below is a separate case.

Ordinary numbers: x can be -2, 0, or 3.5.

Integers: n can be $1,2,3,\ldots$ when counting objects.

Angles: θ can be 30°, 60°, or 90°.

Sets: A can be $\{1,2\}$ or $\{0,3,5\}$.

Greek letters as numbers: λ can be -1, 0, or 2.

Letters do not have permanently assigned types. Uppercase letters often represent sets, and Greek letters often represent angles or numbers, but the context determines their meaning.
Symbols acquire meaning from their mathematical context.